

Day 97 — TNPSC Group 2A Study Material

Part A – General Studies: Current Affairs in Geography of India (8): Noida International Airport (Jewar)

READING MATERIAL

Covers India's newest major airport, Noida International Airport at Jewar — its location, capacity, and 2026 launch timeline.

Q: Where is Noida International Airport located, and what is its IATA code?

A: At Jewar, in the Gautam Buddha Nagar district of Uttar Pradesh — along the Yamuna Expressway, about 72 km south-east of New Delhi. IATA code: DXN.

Q: When was the airport's first phase inaugurated, and when did commercial operations begin?

A: Inaugurated 28 March 2026; commercial operations began 15 June 2026, with an IndiGo flight (6E 2278) from Lucknow as the first flight to land.

Q: What is the airport's first-phase passenger capacity, and its long-term target?

A: 12 million passengers annually in Phase I, rising to 70 million by 2050 (across all phases).

Q: What expansion is planned for Phase III?

A: A second runway (4,150 metres long), Terminal 2, and a Ground Transportation Centre.

Q: Who is developing and operating Noida International Airport, and what is its role for Delhi-NCR?

A: Zurich Airport International AG — serving as a THIRD airport for the Delhi-NCR region, alongside Indira Gandhi International Airport.

MOCK TEST (WITH ANSWERS)

1. Noida International Airport is located at Jewar, in which state?

A. Haryana

B. Uttar Pradesh ✓

C. Delhi

D. Rajasthan

Explanation: Uttar Pradesh (Gautam Buddha Nagar district). [easy, web-research]

2. Noida International Airport's first phase was inaugurated on:

A. 28 January 2026

B. 28 February 2026

C. 28 March 2026 ✓

D. 28 April 2026

Explanation: 28 March 2026. [medium, web-research]

3. Noida International Airport's Phase I passenger capacity is 12 million annually, rising to how much by 2050?

A. 30 million

B. 50 million

C. 70 million ✓

D. 100 million

Explanation: 70 million. [hard, web-research]

4. Who is developing and operating Noida International Airport?

A. Adani Group

B. GMR Group

C. Zurich Airport International AG ✓

D. DIAL

Explanation: Zurich Airport International AG. [hard, web-research]

5. Noida International Airport's IATA code is:

A. NIA

B. JWR

C. DXN ✓

D. NDA

Explanation: DXN. [hard, web-research]

Part B – Aptitude & Mental Ability: Simplification

READING MATERIAL

Simplification is 1 of 10 named Aptitude sub-skills (15 questions total for the whole Aptitude unit). TNPSC's simplification questions are rarely 'plug into one formula' — they combine BODMAS, fractions/decimals, ratios, and algebraic shortcuts. Each item below is a worked method, not just an answer — the goal is to recognize which technique a given expression calls for.

Q: Method — BODMAS with a nested bracket and an unknown: Solve for x : $57 - [42 - \{37 - (20 - x)\}] = 176 \div (8 \div 2)$

A: Work from the innermost bracket outward, and simplify the RHS first: $176 \div (8 \div 2) = 176 \div 4 = 44$. Now the LHS: $(20 - x)$ stays as is for now $\rightarrow \{37 - (20 - x)\} = 17 + x \rightarrow [42 - (17 + x)] = 25 - x \rightarrow 57 - (25 - x) = 32 + x$. Set equal to 44: $32 + x = 44 \rightarrow x = 12$. (Real TNPSC-style problem; this exact expression is a recurring coaching-site reference question.)

Q: Method — sum of two squares without brute-force squaring: Find $217 \times 217 + 183 \times 183$.

A: Use $a^2 + b^2 = (a+b)^2 - 2ab$ instead of squaring two 3-digit numbers by hand. Here $a+b = 400$ (round number — that's the clue this trick applies) and $ab = 217 \times 183 = 39,711$. So $a^2+b^2 = 400^2 - 2(39,711) = 160,000 - 79,422 = 80,578$. Recognizing 'the two numbers add to a round number' is the trigger for this shortcut.

Q: Method — difference of squares shortcut: Simplify $(197 \times 197 - 103 \times 103) \div (197 - 103)$ without long multiplication.

A: $a^2 - b^2 = (a+b)(a-b)$, so $(a^2-b^2)/(a-b) = (a+b)$ directly — no multiplication needed at all. Here $a=197$, $b=103$, so the answer is simply $197+103 = 300$. This 'difference of squares divided by the difference' shape is a favorite TNPSC trick question precisely because it looks like it needs heavy calculation but collapses to one addition.

Q: Method — fraction-of-decimal addition: Simplify $7/5$ of $58 + 3/8$ of 139.2 .

A: $7/5$ of $58 = 7 \times 58/5 = 406/5 = 81.2$. $3/8$ of $139.2 = 3 \times 139.2/8 = 417.6/8 = 52.2$. Sum: $81.2 + 52.2 = 133.4$.

Q: Method — ratio substitution to simplify a compound ratio expression: If $x : y = 3 : 4$, find $(4x + 5y) : (5x - 2y)$.

A: Don't solve for x and y separately — substitute $x=3k$, $y=4k$ (k is just a scaling factor that will cancel). $(4x+5y) = 12k+20k = 32k$. $(5x-2y) = 15k-8k = 7k$. Ratio = $32k : 7k = 32 : 7$. The k always cancels in a ratio question — that's the whole trick.

Q: Method — same ratio-substitution trick, different ratio: If $x : y = 3 : 5$, find $(5x + 3y) : (15x - 2y)$.

A: Substitute $x=3k$, $y=5k$. $(5x+3y) = 15k+15k = 30k$. $(15x-2y) = 45k-10k = 35k$. Ratio = $30k:35k = 6:7$.

Q: Method — simultaneous equations feeding into a simplification target: If $3a + 4b = 22$ and $8a - 5b = -4$, find $(a+b)^2$.

A: Solve the pair normally: from $3a+4b=22$ and $8a-5b=-4$, elimination gives $a=2$, $b=4$ (check: $3(2)+4(4)=6+16=22$ ✓; $8(2)-5(4)=16-20=-4$ ✓). The question doesn't ask for a and b individually — it asks for $(a+b)^2 = (2+4)^2 = 36$. TNPSC often hides the real target $((a+b)^2)$, not a or b at the end of a simultaneous-equation question — always re-read what's actually being asked before solving for individual variables.

Q: Method — mixed numbers with a nested fraction: Simplify $2\frac{1}{2} + 3\frac{1}{3} \div \frac{2}{3}$.

A: BODMAS still applies to fractions — division before addition. $3\frac{1}{3} \div \frac{2}{3} = (10/3) \div (2/3) = (10/3) \times (3/2) = 5$. Then $2\frac{1}{2} + 5 = 7\frac{1}{2}$. The common mistake is adding $2\frac{1}{2}+3\frac{1}{3}$ first — division always binds tighter than addition, fractions or not.

MOCK TEST (WITH ANSWERS)

1. Solve for x : $70 - [45 - \{33 - (15 - x)\}] = 171 \div (9 \div 3)$

- A. 10
- B. 12
- C. 14 ✓**
- D. 16

Explanation: RHS: $171 \div (9 \div 3) = 171 \div 3 = 57$. LHS inner-out: $\{33 - (15 - x)\} = 33 - 15 + x = 18 + x \rightarrow [45 - (18 + x)] = 27 - x \rightarrow 70 - (27 - x) = 43 + x$. Set $43 + x = 57 \rightarrow x = 14$. [medium, authored]

2. Find $312 \times 312 + 288 \times 288$ using the $(a+b)^2 - 2ab$ shortcut.

- A. 179,808
- B. 180,288 ✓**
- C. 190,000
- D. 182,688

Explanation: $a+b=600$, $ab=312 \times 288=89,856$. $a^2+b^2=600^2-2(89,856)=360,000-179,712=180,288$. [medium, authored]

3. Simplify $(425 \times 425 - 375 \times 375) \div (425 - 375)$.

- A. 400
- B. 750
- C. 800 ✓**
- D. 425

Explanation: Using $a^2 - b^2 = (a+b)(a-b)$, dividing by $(a-b)$ leaves just $(a+b) = 425+375 = 800$. [easy, authored]

4. Simplify $\frac{5}{6}$ of 72 + $\frac{2}{7}$ of 98.

- A. 88 ✓**
- B. 84
- C. 90
- D. 86

Explanation: $\frac{5}{6}$ of 72 = 60. $\frac{2}{7}$ of 98 = 28. Sum = 88. [easy, authored]

5. If $p : q = 5 : 6$, find $(3p + 2q) : (4p - q)$.

- A. 27 : 14 ✓**
- B. 9 : 7
- C. 27 : 7
- D. 14 : 9

Explanation: $p=5k, q=6k$. $(3p+2q)=15k+12k=27k$. $(4p-q)=20k-6k=14k$. Ratio = 27:14. [medium, authored]

6. If $m : n = 4 : 7$, find $(2m + 3n) : (5m - n)$.

- A. 29 : 13 ✓**
- B. 13 : 29
- C. 29 : 20
- D. 20 : 29

Explanation: $m=4k, n=7k$. $(2m+3n)=8k+21k=29k$. $(5m-n)=20k-7k=13k$. Ratio = 29:13. [medium, authored]

7. If $5a + 2b = 16$ and $3a - 4b = -6$, find the value of $(a+b)^2$.

- A. 16
- B. 25 ✓**
- C. 36
- D. 9

Explanation: Multiply the first equation by 2: $10a+4b=32$. Add to the second $(3a-4b=-6)$: $13a=26 \rightarrow a=2$. Substitute back: $5(2)+2b=16 \rightarrow b=3$. $(a+b)^2=(2+3)^2=25$. [hard, authored]

8. Simplify $4\frac{2}{3} + 2\frac{1}{4} \div \frac{3}{4}$.

- A. $7\frac{2}{3}$ ✓**
- B. $6\frac{1}{3}$
- C. 7
- D. $6\frac{2}{3}$

Explanation: $2\frac{1}{4} \div \frac{3}{4} = (9/4) \div (3/4) = 3$. Then $4\frac{2}{3} + 3 = 7\frac{2}{3}$. [medium, authored]

9. A trader borrows ₹16,000 at 12% p.a. simple interest and another ₹12,000 at 18% p.a. simple interest. In how many years will the two amounts owed (principal + interest) become equal?

- A. $12\frac{2}{3}$ years
- B. $14\frac{2}{3}$ years
- C. $16\frac{2}{3}$ years ✓**
- D. $18\frac{2}{3}$ years

Explanation: This mirrors a real 2025 TNPSC Aptitude question style (same numbers, Simple Interest unit) — included here as a reminder that 'Simplification' skills (careful fraction/ratio handling) carry directly into other Aptitude sub-topics like Simple Interest. Set $16000+16000(0.12)t = 12000+12000(0.18)t$ and solve for t to get $16\frac{2}{3}$ years. [hard, question-bank-2025]

10. Find the value of: $89 \times 89 + 11 \times 11 + 2 \times 89 \times 11$.

- A. 10,000 ✓**
- B. 9,801
- C. 8,900
- D. 10,201

Explanation: This is the $(a+b)^2$ pattern in disguise: $a^2+2ab+b^2 = (a+b)^2$. Here $a=89$, $b=11$, so the answer is $(89+11)^2 = 100^2 = 10,000$ — no separate multiplication needed. [medium, authored]

Part C – General English: Poem: "The Power of a Smile" — Tupac Shakur

READING MATERIAL

A short poem contrasting destructive emotions (anger, cold/wind) against the healing power of a simple smile. Real coaching-site evidence tests theme and figure-of-speech identification per line.

Q: Who wrote "The Power of a Smile", and what is its central theme?

A: Tupac Shakur. Central theme: the healing power of a smile — contrasted against destructive forces like anger and cold wind.

Q: Which emotion does the poem describe as something that 'tears you apart'?

A: Anger.

Q: Identify the figure of speech: "The power of wind can chill."

A: Personification — wind is given the active, almost intentional power 'to chill', as if performing a deliberate action.

Q: Identify the figure of speech: "The power of a smile... can heal a frozen heart."

A: Metaphor — a 'frozen heart' (emotional coldness/numbness) is described directly, and a smile 'heals' it, without using 'like/as'.

Q: What does "can heal a frozen heart" suggest?

A: That a smile can cure sadness/emotional coldness — restoring warmth and feeling, metaphorically 'unfreezing' a hardened or hurt heart.

MOCK TEST (WITH ANSWERS)

1. "The Power of a Smile" was written by:

A. Langston Hughes

B. Tupac Shakur ✓

C. Maya Angelou

D. W.H. Auden

Explanation: Tupac Shakur. [easy, web-research]

2. The central theme of the poem is:

A. Anger and power

B. Natural forces

C. The healing power of a smile ✓

D. War and peace

Explanation: The healing power of a smile. [easy, web-research]

3. Which emotion is described in the poem as something that 'tears you apart'?

A. Joy

B. Sadness

C. Anger ✓

D. Hope

Explanation: Anger. [medium, web-research]

4. "The power of wind can chill" uses which figure of speech?

A. Simile

B. Metaphor

C. Personification ✓

D. Hyperbole

Explanation: Personification — wind given the active power 'to chill'. [medium, web-research]

5. "The power of a smile... can heal a frozen heart" uses which figure of speech?

A. Metaphor ✓

B. Irony

C. Alliteration

D. Onomatopoeia

Explanation: Metaphor — a 'frozen heart' described directly, without 'like/as'. [medium, web-research]

6. "Can heal a frozen heart" suggests that a smile can:

A. Cause physical damage

B. Cure sadness/emotional coldness ✓

C. Bring summer weather

D. Make the heart literally turn to ice

Explanation: Cure sadness/emotional coldness. [easy, web-research]